

C GOVERNING EQUATIONS IN MIXED-SPECTRAL FORMULATION

In this appendix, we give some more detail to the formulation of the governing equations in mixed-spectral form, as well as their matrix form implemented in PyFuga.

C.1 Recalled equations

The first-order system of equations in Cartesian form is

$$U^0 \frac{\partial u_i^1}{\partial x} + w^1 \frac{\partial U^0}{\partial z} \delta_{i1} = \frac{\partial}{\partial x_j} K \left(\frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right) - \frac{\partial p^1}{\partial x_i} + f_i, \quad (\text{C.1a})$$

$$\frac{\partial u_i^1}{\partial x_i} = 0. \quad (\text{C.1b})$$

The eddy viscosity is modelled as

$$K = \frac{\kappa u_* z}{\phi_m(z/L)}, \quad (\text{C.2})$$

where ϕ_m is the Högström profile function (Högström, 1988). The Monin–Obukhov velocity profile satisfies

$$\frac{\kappa z}{u_*} \frac{\partial U^0}{\partial z} = \phi_m. \quad (\text{C.3})$$

The actuator disc forcing is defined as

$$f_1 = -\frac{1}{2} C_t U_{\text{free}}^2 \delta(x - x_h) \Theta(R^2 - (y - y_h)^2 - (z - z_h)^2), \quad (\text{C.4})$$

where $\Theta(\cdot)$ is the Heaviside step function.

C.2 Fourier-transformed equations

We begin with the first-order momentum equation (C.1a). Since the system is linear, we can apply a Fourier transform in the horizontal plane without introducing convolution in the wavenumber domain; spatial derivatives are replaced with products of the variables and their corresponding wavenumber. Let us analyse each term in the equation. The first term, the convective term, transforms as follows:

$$\mathcal{F}_{(x,y)} \left\{ U^0 \frac{\partial u_i^1}{\partial x} \right\} = U^0 i k_1 \hat{u}_i^1, \quad (\text{C.5})$$

where $\hat{u}_i^1$ denotes the Fourier transform of u_i^1 , and k_1 is the wavenumber in the x -direction. Since U^0 is independent of x and y , it remains a function of z alone in the transformed equations. Here, $i k_1 \hat{u}_i^1$ arises from the Fourier transform of the derivative, which replaces $\frac{\partial}{\partial x}$ with $i k_1$. The pressure gradient term transforms differently depending on the direction i , such that

$$\mathcal{F}_{(x,y)} \left\{ -\frac{\partial p^1}{\partial x_i} \right\} = \begin{cases} -i k_i \hat{p}^1 & \text{for } i = 1 \text{ or } 2, \\ -\frac{\partial p^1}{\partial z} & \text{for } i = 3. \end{cases} \quad (\text{C.6})$$

Note that, in the mixed-spectral formulation, it is often clearer to write out each equation explicitly rather than using Einstein notation, as $i = 1$ and $i = 2$ represent components in wavenumber space, while $i = 3$ remains in physical space.

The forcing term transforms straightforwardly to give

$$\mathcal{F}_{(x,y)} \{ f_i^1 \} = \hat{f}_i^1. \quad (\text{C.7})$$

Next, consider the second term on the left-hand side, the shear term, which transforms to give

$$\mathcal{F}_{(x,y)} \left\{ \frac{\partial U^0}{\partial z} w^1 \delta_{i1} \right\} = \frac{\partial U^0}{\partial z} \hat{w}^1 \delta_{i1}. \quad (\text{C.8})$$

This term remains in the physical domain for z because $\partial U^0/\partial z$ is independent of x and y . The Reynolds stress term is more complex and requires the use of the product rule. It can be written as

$$\begin{aligned} & \mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) \\ &= \mathcal{F}_{(x,y)} \left(\frac{\partial K}{\partial x_j} \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} + K \left\{ \frac{\partial^2 u_i^1}{\partial x_j^2} + \frac{\partial^2 u_j^1}{\partial x_j \partial x_i} \right\} \right). \end{aligned} \quad (\text{C.9})$$

Note that for a simple eddy viscosity closure, $K = K(z)$ and thus cannot be taken outside of $\partial/\partial x_j$. With the equation in this form, we can apply the continuity equation for the first-order perturbation, i.e. divergence-free flow conditions $\partial u_j^1/\partial x_j = 0$, to simplify it to

$$\begin{aligned} & \mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) \\ &= \mathcal{F}_{(x,y)} \left(\frac{\partial K}{\partial x_j} \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} + K \frac{\partial^2 u_i^1}{\partial x_j^2} \right). \end{aligned} \quad (\text{C.10})$$

Expanding the Einstein summation and noting that $K = K(z)$, we can simplify to

$$\mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) = \mathcal{F}_{(x,y)} \left(\frac{dK}{dz} \left\{ \frac{\partial u_i^1}{\partial z} + \frac{\partial w^1}{\partial x_i} \right\} + K \frac{\partial^2 u_i^1}{\partial x_j^2} \right). \quad (\text{C.11})$$

Expanding the last term and applying the Fourier transform, for $i = 1$ or 2 :

$$\begin{aligned} & \mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) \\ &= \frac{\partial K}{\partial z} \left(\frac{\partial \hat{u}_i^1}{\partial z} + ik_i \hat{w}^1 \right) + K \left(-(k_1^2 + k_2^2) \hat{u}_i^1 + \frac{\partial^2 \hat{u}_i^1}{\partial z^2} \right) \\ &= -k^2 K \hat{u}_i^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{u}_i^1}{\partial z} \right) + i \frac{\partial K}{\partial z} k_i \hat{w}^1, \end{aligned} \quad (\text{C.12})$$

with k being the horizontal plane wave vector magnitude. For $i = 3$, we obtain

$$\mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial w^1}{\partial x_j} + \frac{\partial u_j^1}{\partial z} \right\} \right) = -k^2 K \hat{w}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{w}^1}{\partial z} \right) + \frac{\partial K}{\partial z} \frac{\partial \hat{w}^1}{\partial z}. \quad (\text{C.13})$$

Applying the Fourier transform also to the continuity equation (C.1b), we can now write the first-order system of equations in mixed-spectral form as

$$U^0 ik_1 \hat{u}^1 + \frac{\partial U^0}{\partial z} \hat{w}^1 = -k^2 K \hat{u}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{u}^1}{\partial z} \right) + \frac{dK}{dz} ik_1 \hat{w}^1 - ik_1 \hat{p}^1 + \hat{f}_1^1 \quad (\text{C.14a})$$

$$U^0 ik_1 \hat{v}^1 = -k^2 K \hat{v}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{v}^1}{\partial z} \right) + \frac{dK}{dz} ik_2 \hat{w}^1 - ik_2 \hat{p}^1 \quad (\text{C.14b})$$

$$U^0 ik_1 \hat{w}^1 = -k^2 K \hat{w}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{w}^1}{\partial z} \right) + \frac{dK}{dz} \frac{\partial \hat{w}^1}{\partial z} - \frac{\partial \hat{p}^1}{\partial z} \quad (\text{C.14c})$$

$$ik_1 \hat{u}^1 + ik_2 \hat{v}^1 + \frac{\partial \hat{w}^1}{\partial z} = 0 \quad (\text{C.14d})$$

with boundary conditions

$$\hat{u}_i^1(z_0) = 0, \quad \hat{u}_i^1(z_i) = 0. \quad (\text{C.15})$$

Note that the boundary conditions are unaffected by the transformation to the mixed-spectral domain. These boundary conditions are much more conveniently formulated in the spatial domain, which is one of the major reasons for keeping the Cartesian coordinate z in the governing equations. Other reasons include the dependency of K on z , which gives rise to terms that are non-linear in z , such as $K \frac{\partial \hat{w}^1}{\partial z}$, which would be less convenient to solve in the spectral domain.

C.3 Normalisation of the governing equations

To simplify the representation of the horizontal wave vector, we transform from Cartesian coordinates (k_1, k_2) to polar coordinates

$$k_1 = k \cos \beta, \quad k_2 = k \sin \beta, \quad (\text{C.16})$$

where k is the magnitude of the horizontal wavenumber, and β the phase angle. Substituting these definitions into eqs. C.14, we obtain:

$$\begin{aligned} U^0 ik \cos \beta \hat{u}^1 + \frac{\partial U^0}{\partial z} \hat{w}^1 &= -k^2 K \hat{u}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{u}^1}{\partial z} \right) + \frac{\partial K}{\partial z} ik \cos \beta \hat{w}^1 - ik \cos \beta \hat{p}^1 + \hat{f}_1^1 \\ U^0 ik \cos \beta \hat{v}^1 &= -k^2 K \hat{v}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{v}^1}{\partial z} \right) + \frac{\partial K}{\partial z} ik \sin \beta \hat{w}^1 - ik \sin \beta \hat{p}^1 \\ U^0 ik \cos \beta \hat{w}^1 &= -k^2 K \hat{w}^1 + \frac{\partial}{\partial z} (K (-ik \cos \beta \hat{u}^1 - ik \sin \beta \hat{v}^1)) \\ &\quad + \frac{\partial K}{\partial z} (-ik \cos \beta \hat{u}^1 - ik \sin \beta \hat{v}^1) - \frac{\partial \hat{p}^1}{\partial z}, \end{aligned} \quad (\text{C.17})$$

where $\frac{\partial \hat{w}^1}{\partial z}$ has been eliminated from eq. (C.14c) using the continuity equation,

$$ik \cos \beta \hat{u}^1 + ik \sin \beta \hat{v}^1 + \frac{\partial \hat{w}^1}{\partial z} = 0. \quad (\text{C.18})$$

The forcing terms $\hat{f}_i^1(k, z)$ cannot generally be separated into independent functions of k and z . Consider the Fourier transform of the actuator disc (C.4):

$$\begin{aligned} \hat{f}_i(k_x, k_y, z) &= \frac{1}{(2\pi)^2} e^{-ix_h} \int_{-\infty}^{\infty} dy f(x_h, y, z) e^{-iyk_y} \\ &= \frac{1}{(2\pi)^2} e^{-ix_h} \int_{-\infty}^{\infty} dy -\frac{1}{2} C_t U_{\text{free}}^2 \Theta(R^2 - (y - y_h)^2 - (z - z_h)^2) e^{-iyk_y}. \end{aligned} \quad (\text{C.19})$$

The complexity arises because U_{free} is a function of z , multiplied by the Heaviside function which depends on both y and z . This mixed dependence prevents the forcing term from being expressed as a product of independent functions. The forcing term is therefore approximated as a sum of products:

$$\hat{f}_i^1(k, z) = \sum_n f_{n,k}^i \phi_n(z), \quad (\text{C.20})$$

where $f_{n,k}^i$ captures the dependence on k , including details of the thrust coefficient and wind turbine geometry, while $\phi_n(z)$ is a set of triangular basis functions. Moreover, in eq. (C.17) we only show $\hat{f}_1^1(k, z)$, the longitudinal forcing term, because any lateral and vertical components (resulting from yaw or tilt) can be solved independently and then combined to yield the complete solution. This approach is possible due to the linearity of Fuga.

For a forcing applied at a vertical level n , and introducing the non-dimensional variables

$$\tilde{u} = \frac{\hat{u}^1 u_* k}{f_{n,k}^1}, \quad \tilde{v} = \frac{\hat{v}^1 u_* k}{f_{n,k}^1}, \quad \tilde{w} = \frac{\hat{w}^1 u_* k}{f_{n,k}^1}, \quad \tilde{p} = \frac{\hat{p}^1 k}{f_{n,k}^1}, \quad (\text{C.21})$$

together with

$$U^{0*} \equiv \frac{U^0}{u_*} \quad \text{and} \quad K^* \equiv \frac{kK}{u_*}, \quad (\text{C.22})$$

and the non-dimensional vertical coordinate $s = kz$ (with $ds = k dz$), we obtain, after simplification,

$$\begin{aligned}
U^{0*} i \cos \beta \tilde{u} + \frac{\partial U^{0*}}{\partial s} \tilde{w} &= -K^* \tilde{u} + \frac{\partial}{\partial s} \left(K^* \frac{\partial \tilde{u}}{\partial s} \right) + \frac{\partial K^*}{\partial s} i \cos \beta \tilde{w} - i \cos \beta \tilde{p} + \phi_n \\
U^{0*} i \cos \beta \tilde{v} &= -K^* \tilde{v} + \frac{\partial}{\partial s} \left(K^* \frac{\partial \tilde{v}}{\partial s} \right) + \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} - i \sin \beta \tilde{p} \\
U^{0*} i \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{\partial}{\partial s} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) \\
&\quad + \frac{\partial K^*}{\partial s} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) - \frac{\partial \tilde{p}}{\partial s} \\
i \cos \beta \tilde{u} + i \sin \beta \tilde{v} + \frac{\partial \tilde{w}}{\partial s} &= 0.
\end{aligned} \tag{C.23}$$

The last item to address is the simplification of the term $\frac{\partial U^{0*}}{\partial s} \tilde{w}$. From the Monin–Obukhov profile (C.3), substituting $U^0 = u_* U^{0*}$ and $\frac{\partial}{\partial z} = k \frac{\partial}{\partial s}$, one obtains

$$\frac{\partial U^{0*}}{\partial s} = \frac{1}{K^*}. \tag{C.24}$$

Applying this gives the complete non-dimensionalised system:

$$\begin{aligned}
U^{0*} i \cos \beta \tilde{u} + \frac{1}{K^*} \tilde{w} &= -K^* \tilde{u} + \frac{\partial}{\partial s} \left(K^* \frac{\partial \tilde{u}}{\partial s} \right) + \frac{\partial K^*}{\partial s} i \cos \beta \tilde{w} - i \cos \beta \tilde{p} + \phi_n \\
U^{0*} i \cos \beta \tilde{v} &= -K^* \tilde{v} + \frac{\partial}{\partial s} \left(K^* \frac{\partial \tilde{v}}{\partial s} \right) + \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} - i \sin \beta \tilde{p} \\
U^{0*} i \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{\partial}{\partial s} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) \\
&\quad + \frac{\partial K^*}{\partial s} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) - \frac{\partial \tilde{p}}{\partial s} \\
i \cos \beta \tilde{u} + i \sin \beta \tilde{v} + \frac{\partial \tilde{w}}{\partial s} &= 0.
\end{aligned} \tag{C.25}$$

C.4 Governing equations as a system of ODEs

We now rewrite these as a system of ordinary differential equations in six unknowns, $\{\tilde{u}, \tilde{u}', \tilde{v}, \tilde{v}', \tilde{w}, \tilde{p}\}$, where

$$\tilde{u}' = \frac{\partial \tilde{u}}{\partial s} \quad \text{and} \quad \tilde{v}' = \frac{\partial \tilde{v}}{\partial s}. \tag{C.26}$$

From the continuity equation in (C.25),

$$\frac{\partial \tilde{w}}{\partial s} = -i \cos \beta \tilde{u} - i \sin \beta \tilde{v}. \tag{C.27}$$

Substituting the new variables, dividing the u - and v -momentum equations by K^* , and applying the product rule to the diffusion terms yields, for the u - and v -equations:

$$\begin{aligned}
\frac{\partial \tilde{u}'}{\partial s} &= \tilde{u}' + \frac{U^{0*} i \cos \beta \tilde{u}}{K^*} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{u}' + \frac{1}{K^{*2}} \tilde{w} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \cos \beta \tilde{w} + \frac{i \cos \beta \tilde{p}}{K^*} - \frac{\phi_n}{K^*} \\
\frac{\partial \tilde{v}'}{\partial s} &= \tilde{v}' + \frac{U^{0*} i \cos \beta \tilde{v}}{K^*} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{v}' - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} + \frac{i \sin \beta \tilde{p}}{K^*}.
\end{aligned} \tag{C.28}$$

The z -momentum equation is slightly more complex. Focusing on the diffusion terms:

$$\begin{aligned}
&\frac{\partial}{\partial s} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) + \frac{\partial K^*}{\partial s} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) \\
&= -2i \cos \beta \frac{\partial K^*}{\partial s} \tilde{u} - i K^* \cos \beta \tilde{u}' - 2i \sin \beta \frac{\partial K^*}{\partial s} \tilde{v} - i K^* \sin \beta \tilde{v}'.
\end{aligned} \tag{C.29}$$

The full system of ODEs in the s coordinate is then

$$\begin{aligned}
\frac{\partial \tilde{u}}{\partial s} &= \tilde{u}' \\
\frac{\partial \tilde{u}'}{\partial s} &= \left(1 + \frac{i \cos \beta U^{0*}}{K^*}\right) \tilde{u} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{u}' + \left(\frac{1}{K^{*2}} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \cos \beta\right) \tilde{w} + \frac{i \cos \beta}{K^*} \tilde{p} - \frac{\phi_n}{K^*} \\
\frac{\partial \tilde{v}}{\partial s} &= \tilde{v}' \\
\frac{\partial \tilde{v}'}{\partial s} &= \left(1 + \frac{i \cos \beta U^{0*}}{K^*}\right) \tilde{v} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{v}' - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} + \frac{i \sin \beta}{K^*} \tilde{p} \\
\frac{\partial \tilde{w}}{\partial s} &= -i \cos \beta \tilde{u} - i \sin \beta \tilde{v} \\
\frac{\partial \tilde{p}}{\partial s} &= -2i \cos \beta \frac{\partial K^*}{\partial s} \tilde{u} - i K^* \cos \beta \tilde{u}' - 2i \sin \beta \frac{\partial K^*}{\partial s} \tilde{v} - i K^* \sin \beta \tilde{v}' \\
&\quad - (K^* + U^{0*} i \cos \beta) \tilde{w}.
\end{aligned} \tag{C.30}$$

The presence of $K^*(s)$ in the denominator means that these equations become numerically unstable for small values of s (close to the ground). In this case, instead of s we use the independent variable

$$t = U^{0*} \kappa, \quad \text{with} \quad \frac{\partial t}{\partial z} = \frac{\partial U^{0*}}{\partial z} \kappa = \frac{\kappa k}{K^*}. \tag{C.31}$$

The transition point between s and t is defined unambiguously as the point satisfying $\frac{\partial}{\partial s} = \frac{\partial}{\partial t}$ (full details are given in appendix B). The transformation uses

$$\frac{\partial}{\partial s} = \frac{\kappa}{K^*} \frac{\partial}{\partial t}. \tag{C.32}$$

Applying this to (C.25) and defining $\tilde{u}' = \frac{\partial \tilde{u}}{\partial t}$, $\tilde{v}' = \frac{\partial \tilde{v}}{\partial t}$, the system of ODEs in t becomes

$$\begin{aligned}
\frac{\partial \tilde{u}}{\partial t} &= \tilde{u}' \\
\frac{\partial \tilde{u}'}{\partial t} &= \left(\frac{K^{*2}}{\kappa^2} + \frac{i K^* \cos \beta U^{0*}}{\kappa^2}\right) \tilde{u} + \left(\frac{1}{\kappa^2} - \frac{1}{\kappa} \frac{\partial K^*}{\partial t} i \cos \beta\right) \tilde{w} + \frac{i K^* \cos \beta}{\kappa^2} \tilde{p} - \frac{K^* \phi_n}{\kappa^2} \\
\frac{\partial \tilde{v}}{\partial t} &= \tilde{v}' \\
\frac{\partial \tilde{v}'}{\partial t} &= \left(\frac{K^{*2}}{\kappa^2} + \frac{i K^* \cos \beta U^{0*}}{\kappa^2}\right) \tilde{v} - \frac{1}{\kappa} \frac{\partial K^*}{\partial t} i \sin \beta \tilde{w} + \frac{i K^* \sin \beta}{\kappa^2} \tilde{p} \\
\frac{\partial \tilde{w}}{\partial t} &= -i \frac{K^*}{\kappa} \cos \beta \tilde{u} - i \frac{K^*}{\kappa} \sin \beta \tilde{v} \\
\frac{\partial \tilde{p}}{\partial t} &= -2i \cos \beta \frac{\partial K^*}{\partial t} \tilde{u} - i K^* \cos \beta \tilde{u}' - 2i \sin \beta \frac{\partial K^*}{\partial t} \tilde{v} - i K^* \sin \beta \tilde{v}' \\
&\quad - \left(\frac{K^{*2}}{\kappa} + \frac{K^*}{\kappa} U^{0*} i \cos \beta\right) \tilde{w}.
\end{aligned} \tag{C.33}$$

C.5 Matrix formulation of the governing equations

Equations (C.30) and (C.33) define systems that solve for 6 unknowns, $X := \{\tilde{u}, \tilde{u}', \tilde{v}, \tilde{v}', \tilde{w}, \tilde{p}\}$. In matrix form,

$$\frac{\partial X_i}{\partial s} = A_{ij} X_j + F_i \quad (\text{upper layer}), \quad \frac{\partial X_i}{\partial t} = A_{ij} X_j + F_i \quad (\text{near surface}), \tag{C.34}$$

where in the upper layer A is defined by eq. (C.30) as

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 + \frac{U^{0*}}{K^*} i \cos \beta & -\frac{1}{K^*} \frac{\partial K^*}{\partial s} & 0 & 0 & \frac{1}{K^{*2}} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \cos \beta & \frac{1}{K^*} i \cos \beta \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 + \frac{U^{0*}}{K^*} i \cos \beta & -\frac{1}{K^*} \frac{\partial K^*}{\partial s} & -\frac{1}{K^*} \frac{\partial K^*}{\partial s} i \sin \beta & \frac{1}{K^*} i \sin \beta \\ -i \cos \beta & 0 & -i \sin \beta & 0 & 0 & 0 \\ -2i \cos \beta \frac{\partial K^*}{\partial s} & -i K^* \cos \beta & -2i \sin \beta \frac{\partial K^*}{\partial s} & -i K^* \sin \beta & -K^* - U^{0*} i \cos \beta & 0 \end{pmatrix}, \quad (\text{C.35})$$

and F is

$$F = \begin{pmatrix} 0 \\ -\frac{1}{K^*} \phi_n \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad (\text{C.36})$$

and near the surface, A is defined by eq. (C.33) as

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{K^{*2}}{\kappa^2} + \frac{K^* U^{0*}}{\kappa^2} i \cos \beta & 0 & 0 & 0 & \frac{1}{\kappa^2} - \frac{1}{\kappa} \frac{\partial K^*}{\partial t} i \cos \beta & \frac{K^*}{\kappa^2} i \cos \beta \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{K^{*2}}{\kappa^2} + \frac{K^* U^{0*}}{\kappa^2} i \cos \beta & 0 & -\frac{1}{\kappa} \frac{\partial K^*}{\partial t} i \sin \beta & \frac{K^*}{\kappa^2} i \sin \beta \\ -\frac{K^*}{\kappa} i \cos \beta & 0 & -\frac{K^*}{\kappa} i \sin \beta & 0 & 0 & 0 \\ -2i \cos \beta \frac{\partial K^*}{\partial t} & -i K^* \cos \beta & -2i \sin \beta \frac{\partial K^*}{\partial t} & -i K^* \sin \beta & -\frac{K^{*2}}{\kappa} - \frac{K^*}{\kappa} U^{0*} i \cos \beta & 0 \end{pmatrix}, \quad (\text{C.37})$$

and F is

$$F = \begin{pmatrix} 0 \\ -\frac{K^*}{\kappa^2} \phi_n \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (\text{C.38})$$

Note that whilst (C.34) is written as a partial differential equation, this is purely a matter of consistency. Since the equations are solved by parametrising k and β such that s (or t) is the only variable, the partial derivative could equivalently be written as an ordinary derivative. This system of ordinary differential equations can be solved numerically. Details of the solution can be found in appendix D.

REFERENCES

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